The Exner pressure as an additional prognostic variable in the frozen split (dry and moist)

MultiRate notes

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Abstract

Following the pressure-update idea of Jähn and Knoth (poster "Compressible modeling of a cloudy atmosphere using a general pressure evolution equation"), the Exner pressure π is carried as an additional prognostic variable next to the conservative thermodynamic variable. The pressure gradient is written as the product $c_p\hat{\Theta}\,\nabla\pi$, the frozen split of the MIS method integrates π with its own evolution equation in the fast substeps, and π is recomputed from the equation of state after every MIS step (not at the internal stages). The dry formulation (`FV.ExnerModel`, `src/FV/exner.jl`) and the moist formulation with reversible thermodynamics (`FV.MoistExnerModel`, `src/FV/moist.jl`) are described for the flat C-grid finite volumes, together with the variant that carries the pressure p itself (option `pvar = :p`, the standard pressure equation in the form that follows from the conservative variables) and tests: dry warm bubble, gravity-wave channel, moist bubble of Bryan and Fritsch (2002). A moist Exner pressure (moist exponent) is discussed and not recommended.

1 Idea

The thermodynamic variable Θ (dry: $\rho\theta$; moist: $\rho\theta_\rho$) is transported conservatively. The pressure is a function of the state, $p = P(\rho, \Theta, \dots)$; instead of evaluating P in every fast substep, the Exner pressure obeys

$$\partial_t \pi = \frac{\partial \pi}{\partial \rho}\,\partial_t \rho + \frac{\partial \pi}{\partial \Theta}\,\partial_t \Theta + \dots,$$

which keeps $\pi = \pi(\rho, \Theta, \dots)$ exactly in continuous time. The time discretization does not, so π is reset to the equation of state at the end of every MIS step (hook `frozen_step_end!` of `run_frozen`). Within the step all substeps and stages use the prognostic π; the drift $|\pi - \pi(\text{state})|$ before the reset is measured in the tests below.

2 Dry air

With $\Theta = \rho\theta$, $p = p_0(R\Theta/p_0)^\gamma$ and the Exner pressure

$$\pi = \Big(\frac{p}{p_0}\Big)^{R/c_p} = \Big(\frac{R\Theta}{p_0}\Big)^{\gamma-1}, \qquad \nabla p = c_p\,\Theta\,\nabla\pi \quad \text{(exact)}, \qquad \partial_t\pi = c\,\partial_t\Theta, \quad c = \frac{\mathrm{d}\pi}{\mathrm{d}\Theta} = (\gamma-1)\frac{\pi}{\Theta}.$$

The extended system for $(\rho, U, W, \Theta, \pi)$, $M = (U, W)$:

$$\partial_t\rho + \nabla\cdot M = 0, \qquad \partial_t\Theta + \nabla\cdot(\theta M) = 0, \quad \partial_t\pi + c\,\nabla\cdot(\theta M) = 0,$$
$$\partial_t U + \nabla\cdot(Mu) + c_p\Theta\,\partial_x\pi = 0, \quad \partial_t W + \nabla\cdot(Mw) + c_p\Theta\,\partial_z\pi + \rho g = 0.$$

Frozen form. $N(q) = F(q, q)$ with the fast part linear in the evolving z and the coefficients frozen at the stage value y:

$$\rho_t = -\delta_x U - \delta_z W, \qquad \Theta_t = -\delta_x(\theta_x U) - \delta_z(\theta_z W), \qquad \pi_t = -\delta_x(\kappa_x U) - \delta_z(\kappa_z W),$$
$$U_t = -c_p \Theta_x \, \delta_x \pi, \qquad W_t = -c_p \Theta_z \, \delta_z \pi - g\bar{\rho},$$

slow: the momentum advection (shift technique). θ_x, θ_z are the upwind face values of θ, Θ_x, Θ_z the face means of Θ, and κ the product $c\,\theta_f$ of a cell with its face (two coefficients per face, one for each neighbour). Products of frozen quantities are stored as coefficients of their own, because the MIS stage combination $\sum_j w_j C(Y_j)$ must be linear in C. In N the base operator uses the same pressure gradient, `FV_PRESSURE_FORM = :exner`: $c_p \Theta_f \, \delta\pi(\Theta)$.

HEVI. The vertical part is linear in z; eliminating ρ, Θ, π gives per column the tridiagonal system ($r = (c_{\mathrm{CN}}/\Delta z)^2$, $g_g = c_{\mathrm{CN}}^2 g/(2\Delta z)$, $P_k = c_p \Theta_{z,k}$)

$$W_{k-1}\big[-rP_k\kappa^T_{k-1} + g_g\big] + W_k\big[1 + rP_k(\kappa^T_k + \kappa^B_k)\big] + W_{k+1}\big[-rP_k\kappa^B_{k+1} - g_g\big]$$
$$= R^W_k - \tfrac{c_{\mathrm{CN}}}{\Delta z} P_k (R^\pi_k - R^\pi_{k-1}) - \tfrac{c_{\mathrm{CN}} g}{2}(R^\rho_{k-1} + R^\rho_k),$$

κ^B, κ^T the coefficients of the cell below/above the face; the step is Crank–Nicolson.

Balance. For an isentropic atmosphere π is linear in z and the analytic state satisfies the discrete balance $c_p \Theta_f \, \delta_z \pi + g\bar{\rho} = 0$ to round-off, without a background state (residual $6 \cdot 10^{-13}$ against $8 \cdot 10^{-5}$ for the product form $(\mathrm{d}p/\mathrm{d}\Theta)\nabla\Theta$). For a stratified atmosphere the initial column is balanced discretely in the pressure form of the model (ρ_k from the W equation at rest).

3 Moist air

Formulation (I) of the poster, reversible thermodynamics as in Bryan and Fritsch (2002): prognostic ρ, M, $\Theta = \rho\theta_\rho$, $V = \rho_v = \rho q_v$ and $C = \rho_c = \rho q_c$ (poster eqs. 3–5; $q_t = q_v + q_c$ diagnostic),

$$\partial_t \Theta + \nabla \cdot (\theta_\rho M) = Q_\theta, \qquad \partial_t V + \nabla \cdot (q_v M) = -S_c, \qquad \partial_t C + \nabla \cdot (q_c M) = S_c,$$

with the density potential temperature θ_ρ (moist exponent) and the diagnostic pressure

$$p = p_0 \Big(\frac{R_d \, \Theta}{p_0}\Big)^{1/(1-\kappa_M)}, \qquad \kappa_M = \frac{R_m}{c_{pm}},$$

$$R_m = R_d \rho_d + R_v \rho_v, \qquad c_{pm} = c_{pd}\rho_d + c_{pv}\rho_v + c_{pl}\rho_c,$$

$\rho_d = \rho - \rho_v - \rho_c$; the pressure is an explicit function of (ρ, Θ, V, C), without a saturation adjustment. Constants and $e_s(T)$ as in CGDycore.

3.1 Phase change: Fischer–Burmeister relaxation

Phase equilibrium is the complementarity condition $\rho_c \geq 0$, $\rho_{vs}(T) - \rho_v \geq 0$, $\rho_c\,(\rho_{vs} - \rho_v) = 0$, i.e. $\varphi_{FB}(\rho_{vs} - \rho_v, \rho_c) = 0$ with the Fischer–Burmeister function $\varphi_{FB}(a, b) = \frac{1}{2}(a + b - \sqrt{a^2 + b^2})$ (CGDycore `FischerBurmeisterRhoTh`): $\varphi_{FB} < 0$ supersaturated, $\varphi_{FB} > 0$ subsaturated with condensate, and then $\varphi_{FB} \leq \rho_c$. Instead of an adjustment the condensation rate is an explicit relaxation towards $\varphi_{FB} = 0$,

$$S_c = -\frac{\varphi_{FB}(\rho_{vs}(T) - \rho_v,\ \rho_c)}{\tau}, \qquad \tau = \frac{\min(\Delta x, \Delta z)}{w_\tau},$$

with the relaxation time from the grid spacing (a parcel at speed w_τ crosses a cell in τ). Evaporation never removes more than the condensate if $\Delta t \leq \tau$. The latent heating is the prefactor

of Bryan and Fritsch (2002), as CGDycore `SimpleMicrophysicsPot` (there with the evaporation rate $F_{Ph} = -S_c$), with the density-weighted $R_m = R_d\rho_d + R_v\rho_v$, $c_{pm} = c_{pd}\rho_d + c_{pv}\rho_v + c_{pl}\rho_c$, $c_{vm} = c_{pm} - R_m$:

$$Q_\theta = -\Theta\, P\, S_c, \qquad P = -\frac{L}{c_{pm}T} - \ln\frac{p}{p_0}\,\frac{R_m}{c_{pm}}\Big(\frac{R_v}{R_m} + \frac{c_{pl}-c_{pv}}{c_{pm}}\Big) + \frac{R_v}{R_m}.$$

Through the equation of state this is the isochoric first law (internal energy conserved at fixed ρ),

$$\frac{\mathrm{d}T}{\mathrm{d}\rho_c} = \frac{L - R_vT}{c_{vm}}$$

(test, 10^{-6}). The first version kept θ_e (poster eq. 17) fixed instead; that heated 13–38 % too much ($\mathrm{d}T/\mathrm{d}\rho_c = 3631$ against $2738\,\mathrm{K\,m^3/kg}$ at 296 K, 986 hPa), because that θ_e is an isobaric, approximate invariant. Two options: `micro = :slow`, S_c from the stage value y in the slow part $S(y)$ (needs $\tau \gtrsim \Delta t$); `micro = :fast`, S_c from the current fast state in every small step (explicit, needs $\tau \gtrsim \Delta\tau$). The phase change is also a source in the pressure equation (next subsection).

3.2 The simplest product form of the pressure gradient

Use the Exner function with the *dry* exponent,

$$\pi = \Big(\frac{p}{p_0}\Big)^{\kappa_d}, \quad \kappa_d = \frac{R_d}{c_{pd}} \qquad \Longrightarrow \qquad p = p_0\,\pi^{c_{pd}/R_d},$$

$$\nabla p = c_{pd}\,\hat\Theta\,\nabla\pi, \qquad \hat\Theta = \frac{p}{R_d\,\pi} = \frac{p_0^{\kappa_d}\,p^{1-\kappa_d}}{R_d}.$$

This is exact, and the coefficient $\hat\Theta$ is a function of the pressure alone: no moisture variable enters. $\hat\Theta = \rho\,\theta_\rho^{(d)}$ is ρ times the density potential temperature with the dry exponent ($\theta_\rho^{(d)} = T_\rho/\pi$, $T_\rho = p/(\rho R_d)$), i.e. the momentum equation of Bryan and Fritsch, $\nabla p/\rho = c_{pd}\theta_\rho\nabla\pi$. In terms of the prognostic Θ (moist exponent) $\hat\Theta = \Theta\,(p/p_0)^{\kappa_M-\kappa_d}$. With the moist exponent, $\pi_M = (p/p_0)^{\kappa_M}$, the gradient contains $\ln(p/p_0)\,\nabla\kappa_M$ and is not a product. In the frozen form $\hat\Theta_f(y)$ is the face mean of $p/(R_d\pi)$ evaluated with the prognostic π of y.

3.3 Pressure equation

$$\partial_t\pi = a\,\partial_t\rho + b\,\partial_t\Theta + c\,\partial_tV + d\,\partial_tC, \qquad (a,b,c,d) = \frac{\partial\pi}{\partial(\rho,\Theta,V,C)},$$

so the transport part and the phase change both enter. The source of the pressure equation (the Q_p of the poster) follows from $p = R_mT$ with $\mathrm{d}R_m = -R_v\,\mathrm{d}\rho_c$ and the isochoric $\mathrm{d}T$ above:

$$\partial_tp\big|_{\text{phase}} = Q_p = \Big[\frac{R_m(L-R_vT)}{c_{vm}} - R_vT\Big]S_c, \qquad \partial_t\pi\big|_{\text{phase}} = Q_\pi = \kappa_d\,\frac{\pi}{p}\,Q_p,$$

latent heating minus the loss of gas constant. It is identical to the chain rule $b\,(\mathrm{d}\Theta/\mathrm{d}\rho_c) + d - c$ of the equation of state with the Bryan–Fritsch Q_θ (test, 10^{-6}), so π (or p) stays on the equation of state during the step. Frozen per cell (`pressure_phase_factor`), with S_c from y (`:slow`) or from the fast state (`:fast`). At 296 K, 986 hPa: $Q_p/S_c = 7.8\cdot10^5\,\mathrm{Pa\,m^3/kg}$ (first version $1.07\cdot10^6$); a condensation rate of $10^{-5}\,\mathrm{kg\,m^{-3}\,s^{-1}}$ raises p by 8 Pa/s. Without this term p (or π) would drift from the equation of state during the step and be corrected only by the reset. frozen from y (central differences of the explicit equation of state). With the flux divergences of ρ, Θ, V, C the transport part of the π row is $-\delta(\kappa_fF)$ with one coefficient per face and neighbour, $\kappa_f = a + b\,\theta_f + c\,q_{v,f} + d\,q_{c,f}$ (upwind face values); the HEVI column is the dry one with these κ, and V, C follow from its fluxes. With $Q = 0$ the moist model reproduces the dry one (test). After every MIS step π is recomputed from (ρ,Θ,V,C) by the equation of state (no adjustment).

3.4 Initial state of the moist bubble

Saturated, neutral atmosphere $\theta_e = 320\,\mathrm{K}$, $r_t = 0.02$ (θ_e of the poster, eq. 17). The column is in *discrete* hydrostatic balance with the pressure form of the model: p_1 from $\mathrm{d}p/\mathrm{d}z = -g\rho$ up to the first level, then $c_{pd}\hat{\Theta}_f(\pi_k - \pi_{k-1}) + g\Delta z\,\bar{\rho} = 0$ upwards (secant in p_k), each level saturated with $\theta_e = 320\,\mathrm{K}$, $r_t = 0.02$, split into V and C of phase equilibrium ($\varphi_{FB} = 0$ to round-off). The perturbation of θ_ρ by $2\,\mathrm{K}\cos^2(\pi r/2r_c)$, $r_c = 2\,\mathrm{km}$, at fixed pressure and r_t, saturated, as in CGDycore `PerturbMoistProfile`.

4 The pressure itself as the additional variable (`pvar = :p`)

Dry. With $p = P(\Theta)$ the pressure equation follows from the conservative Θ equation,

$$\partial_t p = P'(\Theta)\,\partial_t\Theta = -P'(\Theta)\,\nabla\cdot(\theta M), \qquad P'(\Theta) = \frac{\gamma p}{\Theta} = \frac{c^2}{\theta},$$

and in the continuum this *is* the standard pressure equation:

$$-P'(\Theta)\nabla\cdot(\theta M) = -c^2\nabla\cdot M - \frac{\gamma p}{\theta}\,v\cdot\nabla\theta = -\rho c^2\,\nabla\cdot v - v\cdot\nabla p,$$

since $\nabla p = c^2\nabla\rho + (\gamma p/\theta)\nabla\theta$. It is frozen exactly as the π equation: the p row is $-\delta(\kappa_f F)$ with $\kappa_f = (\gamma p/\Theta)_{\mathrm{cell}}(y)\,\theta_f(y)$ (one coefficient per face and neighbour), the momentum rows have the plain difference $-\delta p$ (no product form needed), the fast part is linear in z, the HEVI column is the tridiagonal system above with $P_k = 1$, and p is reset to $P(\Theta)$ after every MIS step. The frozen coefficient is the derivative $\gamma p/\Theta$ (as in the product form); freezing the secant p/Θ instead was unstable (`docs/FINDINGS.md` §11). The full operator N is the base operator with $\delta p(\Theta)$ (`FV_PRESSURE_FORM = :gradp`).

Moist. $\partial_t p = a\,\partial_t\rho + b\,\partial_t\Theta + c\,\partial_t V + d\,\partial_t C$ with $(a,b,c,d) = \partial p/\partial(\rho,\Theta,V,C)$ plus the phase source Q_p: this is the source term Q_p of the poster (frozen sound speed of the non-equilibrium mixture). The code is the moist model with p in place of π and $-\delta p$ in the momentum rows; the balanced initial column uses $p_k - p_{k-1} + g\Delta z\,\bar{\rho} = 0$.

The advective form $\partial_t p + v\cdot\nabla p = -\rho c^2\nabla\cdot v$ can also be frozen (fast $-(\rho c^2)(y)\nabla\cdot(M_z/\rho_y)$, slow $-v_y\cdot\nabla p_y$, $F(q,q) = N(q)$), but it is not conservative and the pressure advection is in the slow part: p then departs from $P(\Theta)$ by the error of the slow step, and the reset after the step is a real correction, not a cosmetic one. The conservative form above keeps the drift at round-off level times Δt (tables below).

5 Is a moist Exner pressure useful?

A moist Exner function $\pi_M = (p/p_0)^{\kappa_M}$ is not recommended as the prognostic variable:

- The product form is lost: $\nabla\pi_M = \pi_M\,[\kappa_M\nabla\ln p + \ln(p/p_0)\nabla\kappa_M]$, so the pressure gradient needs the correction $-p\,\kappa_M^{-1}\ln(p/p_0)\,\nabla\kappa_M$, which must be discretized consistently with the $\nabla\pi_M$ part or the balance is lost.
- π_M jumps at cloud edges, p does not: κ_M depends on the water content ($\kappa_M = 0.2759$ at the Bryan–Fritsch surface, $\kappa_d = 0.2859$ for dry air). At about 5 km the jump of π_M across a cloud edge from $\Delta\kappa \approx 0.01$ alone is about $5\cdot10^{-3}$, while the hydrostatic change of π over one layer of 250 m is $7.6\cdot10^{-3}$: a spurious signal of the size of the hydrostatic force in $\delta\pi_M$.
- The π_M equation gets a source whenever only κ_M changes (phase changes, without motion), and the frozen coefficient would contain $\nabla\kappa_M$ of the frozen moisture field.

The moist κ_M belongs in the thermodynamic variable (θ_ρ, which makes $p = p_0(R_d\rho\theta_\rho/p_0)^{1/(1-\kappa_M)}$ exact; it changes only through the latent heating Q_θ), the pressure variable keeps the dry exponent (π) or is p itself; they are linked by $\hat{\Theta} = \Theta(p/p_0)^{\kappa_M-\kappa_d} = p/(R_d\pi)$.

6 Results (flat grid, MIS54 with the HEVI fast loop)

Dry warm bubble (`examples/frozen_exner_bubble.jl`): isentropic, 20 km×10 km, 160×80 cells, $\Delta\theta = 2$ K, $T = 1000$ s, errors of θ' against RK4 of the same operator.

Δt (nsub)	2 s (8)	5 s (18)	10 s (36)	15 s (54)
product form	$1.67 \cdot 10^{-2}$	$1.90 \cdot 10^{-2}$	$1.92 \cdot 10^{-2}$	$2.04 \cdot 10^{-1}$
Exner π	$1.72 \cdot 10^{-2}$	$1.93 \cdot 10^{-2}$	$1.98 \cdot 10^{-2}$	$1.99 \cdot 10^{-1}$
pressure p	$1.64 \cdot 10^{-2}$	$1.89 \cdot 10^{-2}$	$1.91 \cdot 10^{-2}$	$2.08 \cdot 10^{-1}$
drift π (rel.)	$5 \cdot 10^{-10}$	$6 \cdot 10^{-9}$	$5 \cdot 10^{-8}$	$1.2 \cdot 10^{-7}$
drift p (rel.)	$1 \cdot 10^{-9}$	$1.2 \cdot 10^{-8}$	$5 \cdot 10^{-8}$	$1.1 \cdot 10^{-7}$

Same solution for all three ($\theta'_{\max} = 1.971$ K, $w_{\max} = 17.35$ m/s), same accuracy and stability (the step to $\Delta t = 15$ s is limited by the advective Courant number ≈ 2 of the slow part), mass and Θ conserved to round-off. Relative drift of the additional variable before the reset at most 10^{-7}. Rest state of the analytic isentropic atmosphere: residual $6 \cdot 10^{-13}$ (π), $6 \cdot 10^{-5}$ (p), $8 \cdot 10^{-5}$ (product form); only π is linear in z.

Gravity-wave channel (`examples/frozen_exner_channel.jl`): Skamarock–Klemp, $N = 0.01$, $U = 20$ m/s, 120×16 cells, discretely balanced columns. Wave error of θ' at $z = 5$ km against RK4 (amplitude $2.3 \cdot 10^{-3}$ K):

Δt [s]	2	5	10	20	50	100	150	200
product form	0.3 %	2.2 %	3.0 %	4.6 %	7.0 %	4.6 %	7.8 %	12 %
Exner π	0.3 %	2.2 %	3.2 %	4.9 %	6.8 %	7.9 %	16 %	24 %
pressure p	0.3 %	2.2 %	3.1 %	4.6 %	6.8 %	4.4 %	7.0 %	8.5 %

Equal up to $\Delta t = 50$ s; at large steps the pressure variable has the smallest error (8.5 % at $\Delta t = 200$ s against 12 % for the product form and 24 % for π). Rest-state residual of the balanced columns $\leq 2 \cdot 10^{-15}$ for all three. Long run (36000 s, $\Delta t = 200$ s): MIS54, MISF53, MIS_FV42 unstable with all three forms, MIS_FV53 stable with all three ($\max|w| = 1.3$, 1.5, $1.4 \cdot 10^{-3}$ m/s for product, π, p).

Long runs and the step limit of MIS54. MIS54 in the long channel (36000 s) with all three forms (pressure reset after every MIS step): stable at $\Delta t = 50$ s ($\max|w| = 1.41$, 1.34, $1.43 \cdot 10^{-3}$ m/s for product, π, p), unstable at $\Delta t = 100$ and 200 s. The product form has no additional variable and fails in the same way, so the limit is that of the MIS method in this channel (vertical Courant number of the slow advection $U\Delta t/\Delta z = 3.2$ at $\Delta t = 100$ s), not the pressure variable.

Divergence damping by extrapolation (`run_frozen(...; smdiv =` β`)`). As in WRF, the momentum part of the fast loop is evaluated at the extrapolated state $\tilde{z} = z + \beta(z - z_{\text{prev}})$, z_{prev} the state of the previous momentum evaluation ($z_{\text{prev}} = z$ at the start of the stage); the momentum rows depend only on the thermodynamic rows, so this extrapolates Θ, π or p (HEVI: the explicit horizontal momentum steps; Verlet: the momentum step, with gravity and sponge). It acts like a divergence damping $\sim \beta c^2 \Delta\tau$. Long channel, $\max|w|$ [m/s] after 36000 s ("growing": far above the stable level $1.5 \cdot 10^{-3}$):

form	method, Δt	$\beta = 0$	0.05	0.1	0.2
product	MIS54, 100 s	unstable	1.2 (growing)	$5.7 \cdot 10^{-3}$	$2.2 \cdot 10^{-3}$
product	MIS54, 200 s	unstable	$1.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-3}$	$1.9 \cdot 10^{-3}$
product	MISF53, 200 s	unstable	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$
π	MIS54, 100 s	unstable	unstable	0.21 (growing)	$2.9 \cdot 10^{-3}$
π	MIS54, 200 s	unstable	$5.2 \cdot 10^{-3}$	$4.9 \cdot 10^{-3}$	$4.5 \cdot 10^{-3}$
π	MISF53, 200 s	unstable	$2.1 \cdot 10^{-3}$	$2.0 \cdot 10^{-3}$	$2.0 \cdot 10^{-3}$
p	MIS54, 100 s	unstable	0.17 (growing)	$2.2 \cdot 10^{-3}$	$2.0 \cdot 10^{-3}$
p	MIS54, 200 s	unstable	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$
p	MISF53, 200 s	unstable	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$
all	MIS_FV42, 200 s	unstable	unstable	unstable	unstable

Wave error at $T = 3000$ s, $\Delta t = 100$ s, MIS54, $\beta = 0/0.05/0.1/0.2$: product 4.6/4.8/5.4/6.8 %, π 7.9/7.6/7.3/7.0 %, p 4.4/4.6/5.3/6.9 %. The extrapolation stabilizes MIS54 and MISF53 at $\Delta t = 200$ s for $\beta \geq 0.05$ (amplitude as with the undamped MIS_FV53, 1.3–$1.5 \cdot 10^{-3}$ m/s) and at $\Delta t = 100$ s for $\beta \geq 0.1$ (product, p) or 0.2 (π); it does not stabilize MIS_FV42. The accuracy cost of $\beta = 0.1$ (the WRF value) is about one percentage point for the product form and p; for π the error even decreases. For this channel: p with $\beta = 0.1$ and MIS54 or MISF53.

Moist bubble (`examples/frozen_moist_bubble.jl`): see Section 8.

7 Stability: extrapolation, number of small steps, MIS without extrapolation

Forward–backward with extrapolated pressure. One acoustic step for a Fourier mode, $u^{n+1} = u^n - \mathrm{i}\,\tilde{p}^n$, $p^{n+1} = p^n - \mathrm{i}\mu^2 u^{n+1}$ (scaled), $\tilde{p}^n = p^n + \beta(p^n - p^{n-1})$, $\mu = c\Delta\tau k$:

$$\lambda^2 - \big(2 - (1+\beta)\mu^2\big)\lambda + (1 - \beta\mu^2) = 0.$$

For $\beta = 0$ the roots lie on the unit circle for $\mu \leq 2$: the fast loop is neutral. With $\beta > 0$ the product of the roots is $1 - \beta\mu^2$, so $|\lambda|^2 = 1 - \beta\mu^2$ for complex roots: every small step damps, the short waves most. Stability needs $\mu \leq 2/\sqrt{1+2\beta}$ (1.83 for $\beta = 0.1$). To leading order this is a divergence damping $\nu\nabla(\nabla \cdot u)$ with

$$\nu = \beta c^2 \Delta\tau = \beta c^2 \, \frac{\Delta t}{n_{\mathrm{sub}}}$$

(Klemp, Skamarock and Ha): at fixed Δt the damping per large step falls like $1/n_{\mathrm{sub}}$, and for $n_{\mathrm{sub}} \to \infty$ the fast loop tends to the exact, undamped fast flow.

Many small steps with extrapolation. Long channel (product form, HEVI, 36000 s), $\max|w|$ [m/s], physical level $1.5 \cdot 10^{-3}$:

method, Δt	n_{sub}	$\beta = 0$	$\beta = 0.1$	$\beta = 0.1 \times n_{\mathrm{sub}}/n_0$
WSRK3, 100 s	18	unstable	$2.6 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$
	36	unstable	$7.2 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$
	72	unstable	0.17	$3.7 \cdot 10^{-2}$
	144	unstable	0.27	–
WSRK3, 200 s	36–288	unstable	unstable	unstable
MIS54, 100 s	18	unstable	$5.7 \cdot 10^{-3}$	$5.7 \cdot 10^{-3}$
	36–144	unstable	$1.5 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$
MIS54, 200 s	36	unstable	$1.8 \cdot 10^{-3}$	$1.8 \cdot 10^{-3}$
	72	unstable	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$
	144	unstable	$4.9 \cdot 10^{-2}$	$1.5 \cdot 10^{-3}$
	288	unstable	unstable	–

(n_0 the smallest n_{sub}; –: $\beta = 0.8$, not run.) With fixed β, more small steps make the run *less* stable, as $\nu \propto 1/n_{\text{sub}}$ predicts; scaling $\beta \propto n_{\text{sub}}$ (fixed ν) restores it as long as $\mu \leq 2/\sqrt{1+2\beta}$ allows. WSRK3 owes its stability entirely to the extrapolation; MIS54 at $\Delta t = 100\,\text{s}$ is stable for all n_{sub} (next paragraphs). The damping should be prescribed as ν, i.e. $\beta = \nu\, n_{\text{sub}}/(c^2\Delta t)$, not as a fixed β.

Without extrapolation. Long channel, $\beta = 0$, $\max|w|$ after 36000 s, $U = 20\,\text{m/s}$, $\Delta x = 2.5\,\text{km}$, $C_{\text{adv}} = U\Delta t/\Delta x$ ($C_{\text{sound}} \approx 17\, C_{\text{adv}}$), n_{sub} at the acoustic limit:

C_{adv} (Δt)	0.1 (12.5 s)	0.2	0.4	0.6	0.8	1.2 (150 s)
WSRK3	unstable	unstable	unstable	unstable	unstable	unstable
MIS_KW3	3.6 (growing)	unstable	unstable	unstable	unstable	unstable
MIS2	stable	stable	stable	stable	$6.2 \cdot 10^{-3}$	unstable
MIS4	stable	stable	stable	stable	$3.6 \cdot 10^{-3}$	$2.3 \cdot 10^{-2}$
MIS4a	stable	stable	stable	stable	unstable	unstable
MIS54, MISF53	stable	stable	stable	unstable	unstable	unstable
MIS_FV42	stable	stable	stable	stable	stable	unstable
MIS_FV53	stable	stable	stable	stable	stable	$1.7 \cdot 10^{-3}$
MIS_DG53	stable	stable	stable	stable	stable	stable

(stable: $\max|w| \leq 1.8 \cdot 10^{-3}$.) Without extrapolation WSRK3 is unstable even at $C_{\text{adv}} = 0.1$, all MIS methods with $\alpha, \gamma \neq 0$ are stable to $C_{\text{adv}} = 0.4$, the designed MIS_FV53 and MIS_DG53 to 1.2.

Why MIS is more stable: damping of the fast modes. In the acoustic-advection test of Knoth and Wensch (2014) the fast (acoustic) and slow (advection) operators commute; per Fourier mode the step with exact fast stages is a scalar $R(\mathrm{i}\omega, \mu)$, $\omega = ck\Delta t$ the acoustic phase per large step, μ the advection eigenvalue times Δt. Expand

$$R(\mathrm{i}\omega, \mu) = e^{\mathrm{i}\omega} \sum_k g_k(\omega)\, \mu^k,$$

($g_k = 1/k!$ for the underlying RK method at $\omega = 0$). Two cases:

- $|g_0(\omega)| = 1$: the step reproduces the fast flow exactly. This holds for WSRK3 (every stage restarts from q^n and integrates over $d_i\Delta t$) and MIS_KW3 (stages chained, $\alpha_{i,i-1} = 1$), and for Lawson and ETD. Then stability for small $\mu = \mathrm{i}\kappa$ is decided by $|R|^2 = 1 + a_1\kappa + a_2\kappa^2 + \ldots$, $a_1 = -2\,\mathrm{Im}\, g_1$, $a_2 = |g_1|^2 - 2\,\mathrm{Re}\, g_2$. WSRK3: $g_1 = \mathrm{sinc}(\omega/2)$ is real ($a_1 = 0$), but $a_2 > 0$ for $\omega > 6.3$ (max 0.11 at $\omega = 8.5$); MIS_KW3: $a_2 > 0$ for $\omega > 2.7$ (up to 6.5); EB4: $a_2 > 0$ for $\omega > 8.6$. The growth $a_2\kappa^2$ exists at arbitrarily small advective Courant numbers and is opposed only by the upwind dissipation $O(C_{\text{adv}}k^4)$ of the advection, and in practice by the extrapolation. Lawson: $g_k = 1/k!$ exactly, $R = e^{\mathrm{i}\omega} R_{RK}(\mu)$, the RK stability on the advection for every C_{sound} (only in this commuting test; with the real split Lawson degrades, Section on RK3, Lawson and ETD).

- $|g_0(\omega)| < 1$: the MIS methods with $\alpha, \gamma \neq 0$. For $S = 0$ stage i starts from $q^n + \sum_j \alpha_{ij}\Delta Y_j$ and is forced by $\sum_j \gamma_{ij}\Delta Y_j/(d_i\Delta t)$; the fast increments themselves enter as initial values and forcing, so the step is not $e^{\Delta t L}$ but a trigonometric approximation of it, exact as $\omega \to 0$ to the order of the method: $1 - |g_0| \sim \omega^4$ (MIS54 ω^6). $|R(\mathrm{i}\omega, 0)|$ at $\omega = 2, 4, 8, 12$: MIS54 0.999, 0.973, 0.578, 0.566; MISF53 0.994, 0.918, 0.538, 0.483; MIS4a 0.990, 0.877, 0.305, 0.663; MIS_FV53 0.987, 0.859, 0.531, 0.313; MIS_DG53 0.994, 0.903, 0.181, 0.220; WSRK3, MIS_KW3, LRK4, EB4 1. This is a built-in, scale-selective damping of the acoustic modes with $ck\Delta t = O(1)$, i.e. of exactly the modes whose resonance with the slow stages makes WSRK3 unstable, while the resolved slow modes ($\omega \ll 1$) are untouched. It depends on

$\omega = ck\Delta t$ only, not on n_{sub}: unlike the extrapolation it does not vanish with many small steps.

Largest stable C_{adv} (resolution 0.05), Knoth–Wensch C-grid, exact fast stages, no extrapolation, for $C_{\mathrm{sound}} = 0, 1, 2, 4, 8, 12, 16$ (0.00: unstable at the smallest C_{adv}):

	0	1	2	4	8	12	16
WSRK3	1.60	1.40	1.30	0.00	0.00	0.00	0.00
MIS_KW3	1.60	1.15	0.65	0.00	0.00	0.00	0.00
MIS2	1.35	0.80	0.50	0.70	1.30	2.00	2.65
MIS4	1.60	1.45	1.30	0.85	1.15	2.00	2.80
MIS4a	1.95	1.75	1.70	1.65	2.05	2.95	0.00
MIS54	2.05	2.10	2.30	2.70	2.35	3.00	2.10
MISF53	2.05	2.20	2.45	2.40	2.55	3.00	0.00
MIS_FV42	1.85	1.65	1.60	1.50	1.20	2.05	2.60
MIS_FV53	2.05	2.15	2.30	2.20	2.85	3.00	2.95
MIS_DG53	2.35	2.35	1.85	1.35	2.15	2.30	3.00
LRK3 / LRK4	1.60	1.60	1.60	1.60	1.60	1.60	1.60/1.70
ETD2	0.85	0.80	0.90	0.00	0.05	0.05	0.10
EB4	1.70	1.50	1.50	1.35	0.00	0.00	0.00

The real FV frozen split (column Fourier blocks with gravity and the vertical discretization, `MethodDesign/app_stability.jl`, exact fast stages, $\Delta x = 2.5\,\mathrm{km}$, $U = 10/20/40\,\mathrm{m/s}$) is more restrictive, because there the fast and slow parts do not commute: largest stable C_{adv} WSRK3 and MIS_KW3 0.0/0.0/0.0, MIS2 0.6/0.6/0.1, MIS4 0.7/0.8/0.2, MIS4a, MIS54, MISF53 0.1/0.2/0.4, MIS_FV42 0.9/0.9/0.7, MIS_FV53 1.2/0.2/0.5, MIS_DG53 1.5/1.5/0.8. The channel runs above are somewhat more stable than these exact-stage limits (finite n_{sub} with the HEVI loop), but they have the same ranking: WSRK3 never, MIS54 to about 0.4, MIS_FV53/MIS_DG53 the furthest.

Moist bubble without extrapolation. 80×40, π, `micro = :fast`, $\tau_{\mathrm{fac}} = 0.25$, 1000 s, $\beta = 0$: WSRK3, MIS54, MISF53, MIS_FV53, MIS_DG53 and EB4 are all stable at $\Delta t = 10$, 20, 30 and 40 s (n_{sub} up to 74, $C_{\mathrm{adv}} = \max|w|\Delta t/\Delta x$ up to 2.2), $\max|w|$ 13.4–13.6 m/s (RK4 13.31), θ_e' error 2.11–2.13 K for all (the unresolved relaxation dominates), total water conserved to $2 \cdot 10^{-15}$. The bubble has no mean wind and lasts 1000 s, too short for the slow growth seen in the 10-hour channel; it does not show the difference between WSRK3 and MIS.

8 Moist bubble results

With the Bryan–Fritsch prefactor (current). 80×40, 1000 s, `micro = :fast`, maximum error of θ_e' [K] against RK4 and $\max|w|$ [m/s] at $\Delta t = 2$, 5, 10 s:

	RK4 ($\max|w|$, $\max\theta_e'$)	MIS54	EB4
π, $\tau_{\mathrm{fac}} = 0.25$	13.31, 2.89	2.11/2.12/2.11 (13.5/13.3/13.4)	2.13/2.13/2.12 (13.3/13.3/13.4)
p, $\tau_{\mathrm{fac}} = 0.25$	13.31, 2.89	2.10/2.12/2.11 (13.8/13.5/13.7)	2.12/2.12/2.11 (13.5/13.5/13.6)
π, $\tau_{\mathrm{fac}} = 4$	12.52, 2.96	0.18/0.14/0.17 (12.5)	0.14/0.14/0.17 (12.5)

WSRK3 at $\tau_{\mathrm{fac}} = 4$: 0.14/0.15/0.16 K. With the energy-consistent heating the bubble is much weaker than with the first version ($\max|w|$ 13.3 against 22.2 m/s, $\theta_e' \le 2.9$ against 6.1 K) and close to the dry 2 K bubble, as intended by Bryan and Fritsch. At $\tau_{\mathrm{fac}} = 0.25$ $\max|w|$ now agrees with RK4 within 4 %, the θ_e' error is still large (the unresolved relaxation); at $\tau_{\mathrm{fac}} = 4$ all schemes agree to 0.2 K. Mass and total water conserved to round-off. *The tables below are from the first version (heating at fixed θ_e, 13–38 % too strong); the comparisons between schemes and between τ remain qualitatively valid.*

Bryan–Fritsch moist bubble, 80×40 cells ($\Delta x = \Delta z = 250\,\mathrm{m}$), $T = 1000\,\mathrm{s}$, MIS54 with HEVI, Fischer–Burmeister relaxation with $w_\tau = 50\,\mathrm{m/s}$ ($\tau = 5\,\mathrm{s}$); reference RK4 of the same operator

(3574 steps). Rest-state residual of the balanced saturated column $7 \cdot 10^{-13}$ (π), $7 \cdot 10^{-13}$ (p). Reference: $\theta'_e \in [-0.69, 6.10]$ K, $\max|w| = 21.0$ m/s. Maximum error of θ'_e [K]:

Δt (nsub)	2 s (4)	5 s (10)	10 s (19)
π, `micro = :slow`	0.133	0.099	0.132
π, `micro = :fast`	0.267	0.229	0.266
p, `micro = :slow`	0.113	0.107	0.120
p, `micro = :fast`	0.229	0.213	0.240

$\max|w|$ 21.0 (`:slow`) and 21.4 m/s (`:fast`); mass and total water $V + C$ conserved to round-off, total Θ grows by $2.4 \cdot 10^{-4}$ (latent heating); relative drift of π or p before the reset $\leq 4 \cdot 10^{-8}$; remaining non-equilibrium $\max|\varphi_{FB}|/\rho = 1.1 \cdot 10^{-4}$ (the lag of the relaxation). π and p give the same solution. `:slow` is stable also at $\Delta t = 10\,\text{s} = 2\tau$ and is about twice as accurate as `:fast` here, and cheaper (13–14 s against 22 s at $\Delta t = 10$ s). The errors hardly depend on Δt: they come from the fast loop (fixed $\Delta\tau$), as in the dry bubble.

Dependence on the relaxation time (40×20 cells, $\Delta x = 500$ m, π, $\Delta t = 5$ s, $\max|w|$ at 1000 s): $\tau = 50$ s ($w_\tau = 10$ m/s) 9.3 m/s, $\tau = 10$ s 19.7, $\tau = 2$ s 23.5, $\tau = 0.5$ s (`:fast`) 25.2 m/s; `:slow` and `:fast` agree to 1 %. `:slow` fails for $\tau = 2\,\text{s} < \Delta t/2$, `:fast` for $\tau = 0.1\,\text{s} < \Delta\tau$. With $\tau = 50$ s the lag leaves a supersaturation of $6 \cdot 10^{-4}$, which also biases the diagnosed θ_e (+1.7 K). Without latent heating ($Q_\theta = 0$, $\rho\theta_\rho$ conserved, the first version) the bubble stalls: Θ/ρ increases by 24 K over 10 km in the saturated neutral background, $\max|w|$ peaks at 3 m/s after 200 s and decays.

Relaxation on the fast scale (default). `micro = :fast` with $\tau = \tau_{\text{fac}}\,\Delta\tau_{\text{ac}}$, $\Delta\tau_{\text{ac}} = 0.85 \cdot 0.92 \min(\Delta x, \Delta z)/c_s$ ($c_s = 350$ m/s, the acoustic step of the fast loop). The default is the shortest stable relaxation, $\tau_{\text{fac}} = 0.25$ ($\tau = 0.14$ s at 250 m, ≈ 0.28 of the actual small step $\Delta t/n_{\text{sub}} = 0.5$ s): 80×40, $\Delta t = 2$, 5, 10 s, π and p, $\max|w| = 26.8$–27.0 m/s at 1000 s. At $\tau_{\text{fac}} = 0.2$ the run does not blow up, but the bubble is wrong ($\max|w| = 8.5$–11.4 m/s at $\Delta t = 2$ and 10 s, 27.1 m/s at 5 s); at $\tau_{\text{fac}} = 0.125$ it is unstable. Stable is not accurate: with $\tau < \Delta\tau$ the relaxation is not resolved by the small step. 80×40, π (the p form is the same within 10 %), maximum error of θ'_e against RK4 with the same τ (RK4 step $\leq \tau/2$):

τ_{fac} (τ)	RK4 $\max|w|$	MIS $\max|w|$ ($\Delta t = 5$ s)	$\Delta t = 2$ s	5 s	10 s
0.25 (0.14 s)	–	26.9	–	–	–
0.5 (0.28 s)	22.2	25.7	2.2	1.8	2.2
1 (0.55 s)	22.1	24.2	1.0	0.86	0.97
2 (1.1 s)	22.0	23.1	0.52	0.41	0.50
4 (2.2 s)	21.7	22.3	0.31	0.27	0.32

The physical solution (RK4) hardly depends on τ in this range (21.7–22.2 m/s; 21.0 at $\tau = 5$ s), the MIS error is the unresolved relaxation and does not decrease with Δt. Refining only the small step at $\tau_{\text{fac}} = 0.5$ ($\Delta t = 5$ s, nsub 10, 20, 40, 80) gives $\max|w| = 25.7$, 23.5, 22.5, 22.3 m/s: convergence to RK4. Recommended: $\tau \gtrsim 2\Delta\tau$ ($\tau_{\text{fac}} = 2$–4), i.e. the shortest relaxation the fast step still resolves.

RK3, Lawson and ETD. Besides MIS the frozen step takes the Wicker–Skamarock RK3 (`WSRK3`, MIS with $\beta = \text{diag}(1/3, 1/2, 1)$), Lawson (`LRK4`, `LRK3`) and ETD (`EB4`, `ETD2`) tableaus. Lawson and ETD freeze the fast part at the start of the step, $F_f(C(q^n), \cdot)$, which is nonlinear in Z (Fischer–Burmeister source with `micro = :fast`); the slow forcing of stage j is the remainder $\tilde{S}(Y_j) = N(Y_j) - F_f(C(q^n), Y_j)$. Lawson propagates the increments with the nonlinear fast

flow, ETD integrates each stage from q^n with the polynomial forcing (evaluated at the substep midpoint). The substeps are $\lceil c\, n_{\mathrm{sub}} \rceil$, so the fast step never exceeds $\Delta t / n_{\mathrm{sub}}$ (with rounding, the half step of Lawson at $n_{\mathrm{sub}} = 5$ had $1.25\Delta\tau$ and the relaxation at its limit broke down). Orders on the toy problem of the test suite (exact fast stages): WSRK3 2; EB4 4, also with the fast coefficient frozen at q^n; ETD2 2; LRK4 and LRK3 only 2, because $\varphi(X + \Delta S) \neq \varphi(X) + e^{\tau L}\Delta S$ for a nonlinear flow.

Bubble, 80×40, π, 1000 s, maximum error of θ'_e [K] against RK4 at $\Delta t = 2$, 5, 10 s:

	MIS54	WSRK3	LRK4	LRK3	EB4	ETD2
$\tau_{\mathrm{fac}} = 0.25$	4.1/3.6/4.0	3.7/3.7/3.9	3.7/3.7/2.6	3.8/3.9/3.6	3.7/3.7/4.0	3.7/3.7/4.0
$\tau_{\mathrm{fac}} = 4$	0.31/0.27/0.32	0.27/0.28/0.29	0.29/0.39/2.3	0.28/1.6/2.6	0.27/0.27/0.29	0.27/0.26/0.37

At the default $\tau_{\mathrm{fac}} = 0.25$ all schemes give the same $\max|w| \approx 27$ m/s (RK4 22.2): the error is the unresolved relaxation in the fast step and does not depend on the slow scheme. With a resolved relaxation ($\tau_{\mathrm{fac}} = 4$, RK4 21.7 m/s) MIS54, WSRK3, EB4 and ETD2 agree within 0.1 K ($\max|w|$ 22.3–22.5 m/s); Lawson degrades with Δt (LRK3 $\max|w| = 38.9$ m/s at $\Delta t = 10$ s). The p form is the same within 10 %. Mass and total water are conserved to round-off by all schemes. Cost per step relative to MIS54 (fast time $\sum d_i$ resp. $\sum c_i$): WSRK3 0.7–0.8, ETD2 0.9, EB4 and LRK4 1.3.

9 Open points

- Relaxation time: the solution depends on τ (Section 8); $\tau \to 0$ is the equilibrium limit, limited by the explicit source ($\tau \gtrsim \Delta t$ for `:slow`, $\tau \gtrsim \Delta\tau$ for `:fast`). A linearly implicit source in the vertical column would remove the limit.
- Larger errors of the dry Exner form in the channel at $\Delta t \geq 100$ s, while the pressure variable is the most accurate: in the π form the frozen face mean Θ_f multiplies the evolving $\delta\pi$, in the p form nothing is frozen in the momentum rows.
- Energy formulation (II) and the combined system (III) of the poster; terrain and cut cells; 3D.